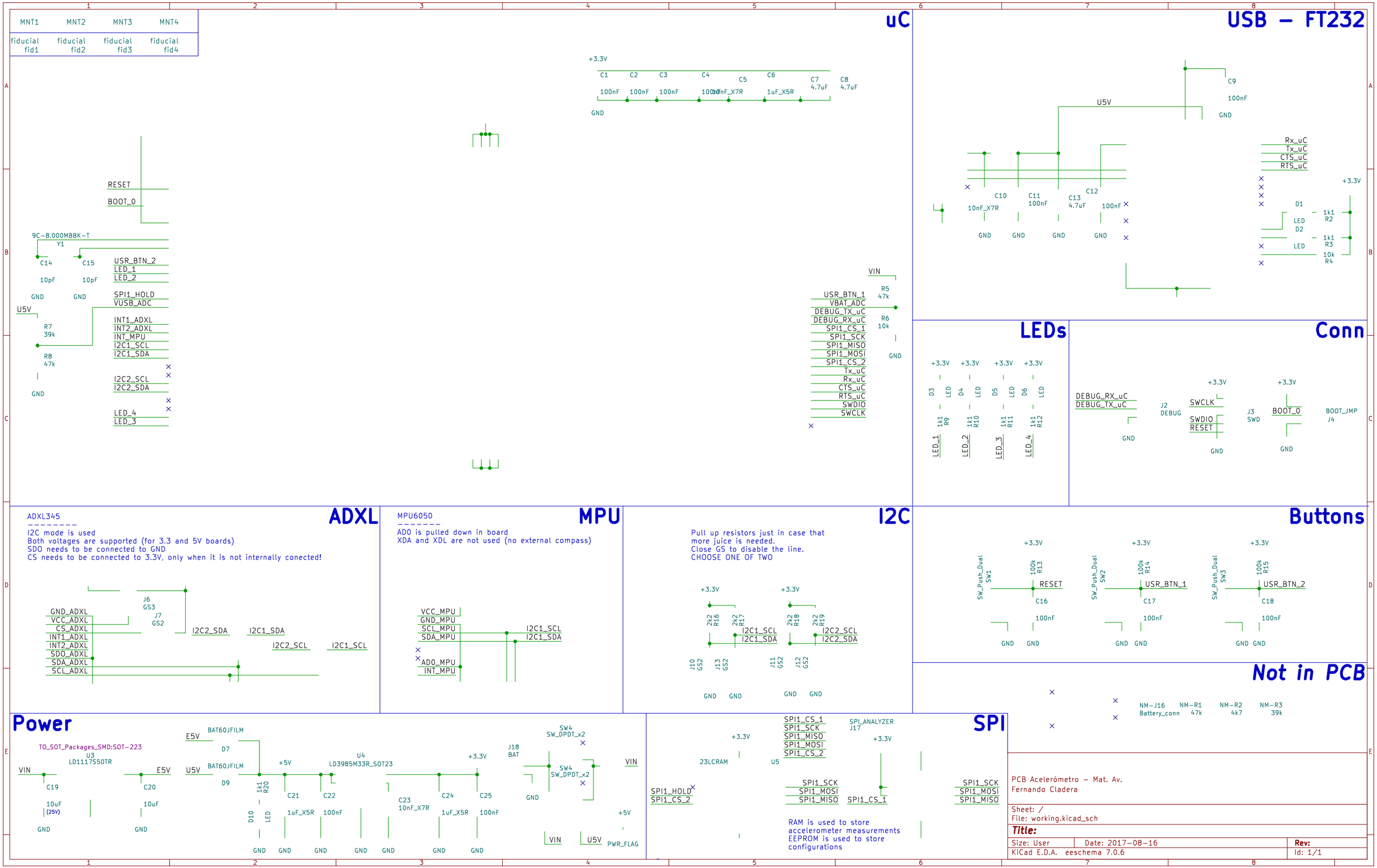

MNT1 MNT2 MNT3 MNT4
fiducial fid1 fiducial fid2 fiducial fid3 fiducial fid4
uC
+3.3V
C1 C2 C3 C4 C5 C6 C7 C8
100nF 100nF 100nF 100nF 10nF_X7R 1uF_X5R 4.7uF 4.7uF
GND
RESET
BOOT_0
9C-8.000MBBK-T
Y1
C14 C15
10pF 10pF
GND GND
USR_BTN_2
LED_1
LED_2
SPI1_HOLD
VUSB_ADC
U5V
R7 39k
R8 47k
INT1_ADXL
INT2_ADXL
INT_MPU
I2C1_SCL
I2C1_SDA
I2C2_SCL
I2C2_SDA
LED_4
LED_3
VIN
R5 47k
R6 10k
USR_BTN_1
VBAT_ADC
DEBUG_TX_uC
DEBUG_RX_uC
SPI1_CS_1
SPI1_SCK
SPI1_MISO
SPI1_MOSI
SPI1_CS_2
Tx_uC
Rx_uC
CTS_uC
RTS_uC
SWDIO
SWCLK
USB – FT232
C9 100nF
U5V
GND
C10 C11 C13 C12
10nF_X7R 100nF 4.7uF 100nF
GND GND GND GND
Rx_uC
Tx_uC
CTS_uC
RTS_uC
+3.3V
D1 D2
LED LED
1k1 R2
1k1 R3
10k R4
LEDs
+3.3V +3.3V +3.3V +3.3V
D3 D4 D5 D6
LED LED LED LED
1k1 R9 1k1 R10 1k1 R11 1k1 R12
LED_1 LED_2 LED_3 LED_4
Conn
DEBUG_RX_uC
DEBUG_TX_uC
GND
J2 DEBUG
+3.3V
SWCLK
SWDIO
RESET
GND
J3 SWD
+3.3V
BOOT_0
GND
BOOT_JMP J4
ADXL
ADXL345
I2C mode is used
Both voltages are supported (for 3.3 and 5V boards)
SDO needs to be connected to GND
CS needs to be connected to 3.3V, only when it is not internally conected!
J6 GS3
J7 GS2
GND_ADXL
VCC_ADXL
CS_ADXL
INT1_ADXL
INT2_ADXL
SDO_ADXL
SDA_ADXL
SCL_ADXL
I2C2_SDA
I2C1_SDA
I2C2_SCL
I2C1_SCL
MPU
MPU6050
AD0 is pulled down in board
XDA and XDL are not used (no external compass)
VCC_MPU
GND_MPU
SCL_MPU
SDA_MPU
AD0_MPU
INT_MPU
I2C1_SCL
I2C1_SDA
I2C
Pull up resistors just in case that
more juice is needed.
Close GS to disable the line.
CHOOSE ONE OF TWO
+3.3V +3.3V
2k2 R16 2k2 R17 2k2 R18 2k2 R19
I2C1_SCL
I2C1_SDA
I2C2_SCL
I2C2_SDA
J10 GS2 J13 GS2 J11 GS2 J12 GS2
GND GND GND GND
Buttons
+3.3V +3.3V +3.3V
SW_Push_Dual SW1 SW_Push_Dual SW2 SW_Push_Dual SW3
100k R13 100k R14 100k R15
RESET USR_BTN_1 USR_BTN_2
C16 C17 C18
100nF 100nF 100nF
GND GND GND GND GND GND
Not in PCB
NM-J16 Battery_conn
NM-R1 47k
NM-R2 4k7
NM-R3 39k
Power
TO_SOT_Packages_SMD:SOT-223
U3 LD1117S50TR
VIN E5V
C19 C20
10uF (25V) 10uF
GND GND
E5V BAT60JFILM D7
U5V BAT60JFILM D9
+5V
U4 LD3985M33R_SOT23
1k1 R20
D10 LED
C21 C22
1uF_X5R 100nF
C23 10nF_X7R
C24 C25
1uF_X5R 100nF
+3.3V
J18 BAT
SW4 SW_DPDT_x2
SW4 SW_DPDT_x2
GND
VIN
+5V
VIN U5V PWR_FLAG
GND GND GND GND GND GND GND
SPI
+3.3V
23LCRAM
U5
SPI1_HOLD
SPI1_CS_2
SPI1_CS_1
SPI1_SCK
SPI1_MISO
SPI1_MOSI
SPI1_CS_2
SPI_ANALYZER J17
+3.3V
SPI1_SCK
SPI1_MOSI
SPI1_MISO
SPI1_CS_1
SPI1_SCK
SPI1_MOSI
SPI1_MISO
RAM is used to store
accelerometer measurements
EEPROM is used to store
configurations
PCB Acelerómetro – Mat. Av.
Fernando Cladera
Sheet: /
File: working.kicad_sch
Title:
Size: User
Date: 2017-08-16
Rev:
KiCad E.D.A. eeschema 7.0.6
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